

Full-syllabus CSE practice paper

200 questions · 200 marks · suggested time 240 minutes

Difficulty is pitched at CIL MT Paper II level and above. Roughly 55% factual/theory and 45% solving (numerical, output prediction, trace-and-count). Every core subject plus OOP, Java, C++, Python, AI/ML and Software Engineering.

Marking for practice purposes: +1 correct, -0.25 wrong (IOCL/IBPS convention). Score yourself both with and without negative marking so you can compare against your CIL result.

Answer key with explanations is at the end. Do not scroll until you have finished.

Section	Subject	Questions
A	Computer Organization & Architecture / Digital Logic	1–22
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Software Engineering questions are distributed inside Section J.

Section A — Computer Organization & Architecture / Digital Logic (Q1-22)

- Q1.** A machine uses 32-bit addresses and is word-addressable with a word size of 4 bytes. What is the maximum size of physical memory it can address? A. 4 GB B. 8 GB C. 16 GB D. 32 GB
- Q2.** The 32-bit IEEE 754 single-precision number 0100 0001 1100 1000 0000 0000 0000 0000 represents which decimal value? A. 12.5 B. 18.0 C. 25.0 D. 50.0
- Q3.** A direct-mapped cache of size 16 KB uses a block size of 32 bytes on a machine with 32-bit addresses. How many bits are used for the tag field? A. 14 B. 16 C. 18 D. 20
- Q4.** How many $128K \times 8$ RAM chips are required to construct a $1M \times 16$ memory? A. 8 B. 12 C. 16 D. 32
- Q5.** In an associative (content-addressable) memory, the register that holds the mask specifying which fields of the argument participate in the comparison is the: A. Argument register B. Key register C. Match register D. Memory buffer register
- Q6.** A disk rotates at 12,000 RPM. What is its average rotational latency? A. 2.0 ms B. 2.5 ms C. 5.0 ms D. 10.0 ms
- Q7.** A RAID 5 array is built from 6 disks of 2 TB each. What is the usable storage capacity? A. 6 TB B. 8 TB C. 10 TB D. 12 TB
- Q8.** A 5-stage pipeline has a clock period of 2 ns. Assuming no stalls, how long does it take to execute 1000 instructions? A. 2000 ns B. 2008 ns C. 2010 ns D. 5000 ns
- Q9.** Memory access takes 100 ns and cache access takes 10 ns. If the cache hit ratio is 0.9 and the cache is accessed first (hierarchical access), the effective memory access time is: A. 19 ns B. 20 ns C. 28 ns D. 100 ns
- Q10.** How many parity (check) bits are required to build a single-error-correcting Hamming code for 8 data bits? A. 3 B. 4 C. 5 D. 8
- Q11.** The binary number 11011011 is equivalent to which of the following? A. $(DB)_{16}$ and $(333)_8$ B. $(BD)_{16}$ and $(333)_8$ C. $(DB)_{16}$ and $(322)_8$ D. $(BC)_{16}$ and $(331)_8$
- Q12.** In 8-bit two's complement representation, the range of representable integers is: A. -127 to +127 B. -128 to +127 C. -128 to +128 D. 0 to 255
- Q13.** The minimum number of flip-flops required to construct a MOD-12 counter is: A. 3 B. 4 C. 6 D. 12
- Q14.** In a JK flip-flop, when $J = 1$ and $K = 1$, the output: A. Is set to 1 B. Is reset to 0 C. Toggles D. Remains unchanged
- Q15.** Which addressing mode is best suited for accessing successive elements of an array inside a loop? A. Immediate B. Direct C. Indexed D. Register direct

- Q16.** During DMA transfer, "cycle stealing" refers to: A. The DMA controller taking the bus for one cycle at a time between CPU cycles B. The CPU halting entirely until the transfer completes C. The DMA controller interrupting after every byte D. Data being copied through the accumulator
- Q17.** In a stack-based (zero-address) machine, the expression $\boxed{AB+CD-*}$ corresponds to which infix expression? A. $(A+B)(C-D)$ B. $A+BC-D$ C. $(A+B)-(CD)$ D. $A(B+C)-D$
- Q18.** Which sequence correctly orders the memory hierarchy from fastest to slowest? A. Cache $\rightarrow$ Registers $\rightarrow$ Main memory $\rightarrow$ Secondary storage B. Registers $\rightarrow$ Cache $\rightarrow$ Main memory $\rightarrow$ Secondary storage C. Registers $\rightarrow$ Main memory $\rightarrow$ Cache $\rightarrow$ Secondary storage D. Main memory $\rightarrow$ Cache $\rightarrow$ Registers $\rightarrow$ Secondary storage
- Q19.** Which statement about SRAM and DRAM is correct? A. DRAM is faster and requires no refresh B. SRAM stores a bit in a capacitor and needs periodic refresh C. SRAM is faster and costlier per bit; DRAM needs periodic refresh D. Both have identical access times but differ only in cost
- Q20.** A machine has 64 KB of main memory and an instruction format with a single memory address field. What is the minimum number of bits needed for the address field? A. 8 B. 12 C. 16 D. 32
- Q21.** The Boolean expression $\boxed{F = A'B + AB' + AB}$ simplifies to: A. $A + B$ B. AB C. $A'B'$ D. $A \oplus B$
- Q22.** In a two-level cache system, L1 hit time is 2 ns with a 95% hit rate, L2 hit time is 10 ns with a 90% hit rate (of L1 misses), and main memory access is 100 ns. What is the effective access time? A. 3.0 ns B. 3.4 ns C. 4.2 ns D. 5.0 ns
- ## Section B — Operating Systems (Q23-42)
- Q23.** Consider the reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. Using FIFO page replacement with **3** frames, how many page faults occur? A. 7 B. 9 C. 10 D. 12
- Q24.** For the same reference string as Q23, using FIFO with **4** frames, the number of page faults is: A. 8 B. 9 C. 10 D. 11
- Q25.** The result of Q23 and Q24 together demonstrates: A. The working set model B. Belady's anomaly C. Thrashing D. The principle of locality
- Q26.** For the reference string of Q23 with 3 frames, LRU replacement produces how many page faults? A. 8 B. 9 C. 10 D. 12
- Q27.** A system uses 32-bit virtual addresses with a 4 KB page size and 4-byte page table entries. The size of a single-level page table for one process is: A. 1 MB B. 2 MB C. 4 MB D. 8 MB
- Q28.** TLB access takes 20 ns and main memory access takes 100 ns. With a TLB hit ratio of 80%, the effective memory access time (page table lookup requires one extra memory access) is: A. 120 ns B. 136 ns C. 140 ns D. 160 ns

Q29. Four processes arrive at time 0 with burst times 6, 8, 7 and 3. Under Shortest Job First scheduling, the average waiting time is: A. 6.25 B. 7.00 C. 7.75 D. 8.50

Q30. How many child processes are created by the following code?

```
☐
c
fork(); fork(); fork();
```

A. 3 B. 6 C. 7 D. 8

Q31. Which of the following is **not** one of the four necessary conditions for deadlock? A. Mutual exclusion B. Hold and wait C. Preemption D. Circular wait

Q32. The disk head is at cylinder 53. Pending requests are 98, 183, 37, 122, 14, 124, 65, 67. Using SSTF, what is the total head movement? A. 208 B. 236 C. 640 D. 236 + 53

Q33. Fixed partitioning of main memory primarily suffers from: A. External fragmentation only B. Internal fragmentation C. Thrashing D. Belady's anomaly

Q34. Thrashing occurs when: A. The page fault rate is so high that the CPU spends more time paging than executing B. Too many processes terminate simultaneously C. The page table exceeds the size of main memory D. Disk scheduling starves distant requests

Q35. Which page replacement algorithm can never suffer from Belady's anomaly? A. FIFO B. LRU C. Second chance D. Clock

Q36. The `read()` system call: A. Runs entirely in user mode for speed B. Copies data from kernel space to user space, possibly after reading device buffers C. Copies data directly from device hardware to user space with no kernel involvement D. Always blocks until end of file is reached

Q37. In the Banker's algorithm, a state is safe if: A. No process is currently waiting B. There exists a sequence in which every process can obtain its maximum need and terminate C. The available vector is all zeros D. Every process holds fewer resources than it requested

Q38. Which of the following is **not** a valid file allocation method? A. Contiguous B. Linked C. Indexed D. Associative

Q39. In a preemptive priority scheduling algorithm, when a higher-priority process arrives in the ready queue: A. It waits until the current process finishes its burst B. The running process is preempted and moved back to the ready queue C. The running process is terminated D. Priorities are recomputed for all processes

Q40. Segmentation differs from paging primarily because: A. Segments are fixed size while pages vary B. Segments are variable-length logical units while pages are fixed-size physical blocks C. Segmentation causes internal fragmentation while paging causes external D. Segmentation requires no address translation

Q41. A binary semaphore initialised to 1 and used around a critical section provides: A. Bounded waiting only B. Mutual exclusion C. Deadlock freedom guarantee D. Priority inversion

Q42. Which of the following transitions is **impossible** in a standard process state diagram? A. Running → Ready B. Ready → Running C. Blocked → Running D. Running → Blocked

Section C — Computer Networks (Q43–60)

Q43. The network 192.168.10.0/26 is subnetted. How many usable host addresses are available per subnet? A. 30 B. 62 C. 64 D. 126

Q44. For the address block 192.168.10.0/26, the broadcast address of the **second** subnet is: A. 192.168.10.63 B. 192.168.10.127 C. 192.168.10.128 D. 192.168.10.191

Q45. A channel has a bandwidth of 3000 Hz and a signal-to-noise ratio of 1023. According to Shannon's theorem, the maximum data rate is: A. 3000 bps B. 6000 bps C. 30,000 bps D. 60,000 bps

Q46. In Go-Back-N ARQ with an n-bit sequence number field, the maximum sender window size is: A. 2^n B. $2^n - 1$ C. 2^{n-1} D. $n - 1$

Q47. In Selective Repeat ARQ with an n-bit sequence number field, the maximum window size is: A. 2^n B. $2^n - 1$ C. 2^{n-1} D. n

Q48. Which OSI layer is responsible for end-to-end error recovery and flow control between processes? A. Data link B. Network C. Transport D. Session

Q49. In the DNS hierarchy, the root name server's primary function is to: A. Resolve the complete domain name to an IP address B. Direct the query to the appropriate top-level domain server C. Cache all domain records globally D. Perform reverse DNS lookups only

Q50. The minimum frame size in Ethernet (CSMA/CD) exists in order to: A. Reduce header overhead B. Ensure a collision is detected before transmission of the frame completes C. Match the MTU of IP D. Simplify CRC computation

Q51. Which pair of port numbers is correct? A. HTTP 8080, HTTPS 443 B. SMTP 25, DNS 53 C. FTP 22, SSH 21 D. Telnet 25, POP3 53

Q52. A 1 Gbps link has a one-way propagation delay of 10 ms. The bandwidth-delay product is: A. 1.25 MB B. 100 Mb C. 100 Mb D. 1 Gb

Q53. Which statement correctly distinguishes connection-oriented from connectionless service? A. Connectionless has higher overhead than connection-oriented B. Connection-oriented establishes a path and guarantees ordered, reliable delivery; connectionless does not C. Both guarantee retransmission of lost segments D. Connection-oriented requires no acknowledgements

Q54. A router operates primarily at which OSI layer? A. Physical B. Data link C. Network D. Transport

Q55. During TCP slow start, the congestion window: A. Increases linearly by one MSS per RTT B. Doubles every RTT until the slow-start threshold is reached C. Remains constant until a timeout D. Halves on every

acknowledgement

Q56. An IPv6 address is how many bits long? A. 32 B. 64 C. 128 D. 256

Q57. The protocol used to map a known IP address to a physical MAC address is: A. ARP B. RARP C. ICMP D. DHCP

Q58. Which routing algorithm requires each router to maintain complete topology information about the entire network? A. Distance vector B. Link state C. Flooding D. Hierarchical

Q59. A frame of 1500 bytes is sent over a 10 Mbps link. The transmission time is: A. 0.12 ms B. 1.2 ms C. 12 ms D. 150 ms

Q60. Which of the following best reflects a layered-design issue that applies at multiple layers of a network architecture? A. Only the physical layer is concerned with transmission media B. Addressing, error control, flow control and sequencing must each be considered at multiple layers C. Once error control is done at the data link layer, no upper layer needs it D. Addressing is handled exclusively by the network layer

Section D — Data Structures & Algorithms (Q61–80)

Q61. A node in a singly linked list typically contains: A. Data only B. A pointer only C. Data and one pointer D. Data and two pointers

Q62. In a circular queue implemented with an array of capacity n , the queue is full when: A. $\text{rear} == \text{front}$ B. $(\text{rear} + 1) \% n == \text{front}$ C. $\text{rear} == n - 1$ D. $\text{front} == 0$ and $\text{rear} == n$

Q63. Which traversal, when combined with **preorder**, uniquely reconstructs a binary tree? A. Postorder B. Level order C. Inorder D. Preorder alone suffices

Q64. The number of distinct binary search trees that can be formed with 4 distinct keys is: A. 8 B. 14 C. 16 D. 24

Q65. The recurrence $T(n) = 2T(n/2) + n$ solves to: A. $\Theta(n)$ B. $\Theta(n \log n)$ C. $\Theta(n^2)$ D. $\Theta(\log n)$

Q66. The time complexity of building a binary heap from an unsorted array of n elements using heapify is: A. $O(1)$ B. $O(\log n)$ C. $O(n)$ D. $O(n \log n)$

Q67. In a binary heap with n nodes stored in an array, the number of leaf nodes is: A. $\lceil n/2 \rceil$ B. $\lfloor n/2 \rfloor$ C. $n/4$ D. $\log n$

Q68. Quicksort exhibits its worst-case time complexity of $O(n^2)$ when: A. The array is randomly ordered B. The pivot always splits the array into two equal halves C. The pivot is consistently the smallest or largest element D. All elements are distinct

Q69. Which sorting algorithm is stable **and** has $O(n \log n)$ worst-case time complexity? A. Quicksort B. Heapsort C. Merge sort D. Selection sort

- Q70.** Evaluate the postfix expression $\boxed{5\ 3\ +\ 8\ 2\ -\ *}$. A. 24 B. 40 C. 48 D. 64
- Q71.** In a complete undirected graph with 8 vertices, the number of edges is: A. 16 B. 28 C. 56 D. 64
- Q72.** Dijkstra's shortest path algorithm fails when the graph contains: A. Cycles B. Negative edge weights C. Disconnected components D. More than 1000 vertices
- Q73.** The best approach to delete an element from an **unsorted** array in $O(1)$ time is: A. Shift all subsequent elements left B. Replace the element with the last element and decrement the size C. Sort the array first, then delete D. Replace the element with 0
- Q74.** Which data structure is most appropriate for implementing an LRU cache with $O(1)$ get and put? A. Array with binary search B. Hash map combined with a doubly linked list C. Min-heap D. Binary search tree
- Q75.** The maximum number of comparisons required by binary search on a sorted array of 1000 elements is: A. 9 B. 10 C. 100 D. 500
- Q76.** The minimum number of nodes in an AVL tree of height 4 (counting a single node as height 0) is: A. 7 B. 9 C. 12 D. 15
- Q77.** Which statement about insertion sort is correct? A. It is $O(n \log n)$ in the worst case but unstable B. It is adaptive and runs in $O(n)$ on nearly sorted input, but $O(n^2)$ in the worst case C. It requires $O(n)$ auxiliary space D. It performs identically on sorted and reverse-sorted input
- Q78.** In separate chaining hash tables, the average time complexity of a search with load factor α is: A. $O(1)$ B. $O(1 + \alpha)$ C. $O(\alpha^2)$ D. $O(\log n)$
- Q79.** A sparse matrix representation is **least** beneficial for: A. A diagonal matrix B. An adjacency matrix of a sparse graph C. A large upper-triangular matrix D. A matrix in which most elements are non-zero
- Q80.** Which technique does the standard 0/1 knapsack solution use? A. Greedy B. Divide and conquer C. Dynamic programming D. Backtracking only

Section E — Database Management Systems (Q81–98)

- Q81.** Each row of a relational table represents a: A. Tuple B. Attribute C. Domain D. Column
- Q82.** The entity integrity constraint ensures that: A. No foreign key value can be NULL B. No primary key value (or any part of it) can be NULL C. All attributes belong to the same domain D. Every tuple references another tuple
- Q83.** In an ER diagram, an entity set is represented by a: A. Diamond B. Ellipse C. Rectangle D. Triangle
- Q84.** Which statement about the PROJECT operation in relational algebra is correct? A. It is a binary operation that removes tuples B. It is a unary operation that removes attributes, producing a vertical subset C. It produces a horizontal partition of the relation D. It always preserves the number of tuples

- Q85.** A relation $R(A, B, C, D)$ has functional dependencies $A \rightarrow B$, $B \rightarrow C$ and $C \rightarrow D$. The relation is in: A. 1NF only B. 2NF but not 3NF C. 3NF but not BCNF D. BCNF
- Q86.** A banking transaction commits successfully and the system crashes immediately afterwards. Which mechanism guarantees that the committed change survives? A. Two-phase locking B. Redo log with write-ahead logging C. Transaction isolation D. Cascading rollback
- Q87.** Which isolation level permits phantom reads but prevents dirty and non-repeatable reads? A. Read uncommitted B. Read committed C. Repeatable read D. Serializable
- Q88.** An SQL schema is best described as: A. A single table holding all user data and constraints B. A named collection of database objects with an authorization identifier for its owner C. A view aggregating multiple tables D. A mechanism that automatically enforces all foreign keys
- Q89.** Which of the following is a DDL command? A. UPDATE B. INSERT C. ALTER D. SELECT
- Q90.** Consider two tables A (10 rows) and B (5 rows). The number of rows produced by `SELECT * FROM A CROSS JOIN B` is: A. 5 B. 15 C. 50 D. 10
- Q91.** Which normal form removes transitive dependency of non-prime attributes on the primary key? A. 1NF B. 2NF C. 3NF D. BCNF
- Q92.** In a B+ tree, unlike a B-tree: A. Keys are stored only in internal nodes B. All data pointers reside in leaf nodes, which are linked C. Height grows linearly with the number of records D. Duplicate keys are prohibited
- Q93.** Ensuring the isolation property of transactions is the responsibility of the: A. Recovery manager B. Concurrency control component C. Query optimiser D. Buffer manager
- Q94.** Which SQL clause filters groups after aggregation? A. WHERE B. HAVING C. GROUP BY D. ORDER BY
- Q95.** A candidate key that is not chosen as the primary key is called: A. Super key B. Foreign key C. Alternate key D. Composite key
- Q96.** Two-phase locking (2PL) guarantees: A. Deadlock freedom B. Conflict serializability C. Recoverability D. Absence of starvation
- Q97.** In RAID 1 built from 8 disks of 1 TB each, the usable capacity and fault tolerance are: A. 8 TB, no tolerance B. 4 TB, survives one failure per mirrored pair C. 7 TB, survives one failure D. 6 TB, survives two failures
- Q98.** A relation is in BCNF if, for every non-trivial functional dependency $X \rightarrow Y$: A. Y is a prime attribute B. X is a super key C. X is a candidate key only D. Y is fully dependent on X

Section F — Theory of Computation (Q99–110)

- Q99.** The minimum number of states in a DFA over $\{a, b\}$ that accepts all strings with an even number of a's is: A. 1 B. 2 C. 3 D. 4

- Q100.** Which regular expression describes all strings over $\{a, b\}$ containing an even number of a's? A. $(babab^*)^*$ B. $(bab)^*$ C. $(a+b)a(a+b)$ D. ab
- Q101.** A system must recognise **one or more** consecutive occurrences of 'a'. The correct regular expression is: A. a^* B. a^+ C. $a?$ D. ab^*
- Q102.** The minimum number of states in a DFA that accepts binary strings representing numbers divisible by 3 is: A. 2 B. 3 C. 4 D. 8
- Q103.** Converting an NFA with n states to an equivalent DFA can require, in the worst case: A. n states B. $2n$ states C. n^2 states D. 2^n states
- Q104.** The language $L = \{a^n b^n \mid n \geq 1\}$ is: A. Regular B. Context-free but not regular C. Context-sensitive but not context-free D. Not recursively enumerable
- Q105.** A Mealy machine produces output based on: A. The current state only B. The current input symbol only C. Both the current state and the current input D. The next state only
- Q106.** A Moore machine with n states, when converted to an equivalent Mealy machine, requires: A. At most n states B. Exactly $2n$ states C. At least n^2 states D. $n + 1$ states
- Q107.** Which class of automaton is required to recognise a context-free language? A. Finite automaton B. Pushdown automaton C. Linear bounded automaton D. Turing machine only
- Q108.** In the Chomsky hierarchy, Type 1 grammars generate: A. Regular languages B. Context-free languages C. Context-sensitive languages D. Recursively enumerable languages
- Q109.** Regular languages are closed under all of the following **except**: A. Union B. Intersection C. Complement D. None — they are closed under all three
- Q110.** The halting problem for Turing machines is: A. Decidable B. Undecidable but recursively enumerable C. Not recursively enumerable D. Decidable only for deterministic machines

Section G — Compiler Design (Q111–122)

- Q111.** The correct order of compiler phases is: A. Lexical $\rightarrow$ syntax $\rightarrow$ semantic $\rightarrow$ intermediate code $\rightarrow$ optimisation $\rightarrow$ code generation B. Syntax $\rightarrow$ lexical $\rightarrow$ semantic $\rightarrow$ code generation $\rightarrow$ optimisation C. Lexical $\rightarrow$ semantic $\rightarrow$ syntax $\rightarrow$ optimisation $\rightarrow$ code generation D. Semantic $\rightarrow$ lexical $\rightarrow$ syntax $\rightarrow$ intermediate code $\rightarrow$ optimisation
- Q112.** Which of the following is **not** a token produced by a lexical analyser? A. Keyword B. Identifier C. Parser D. Operator
- Q113.** A grammar containing left recursion cannot be directly used by: A. LR parsers B. Predictive (LL(1)) parsers C. Operator precedence parsers D. SLR parsers

- Q114.** The four possible actions of a shift-reduce parser are: A. Shift, jump, reduce, halt B. Reduce, copy, accept, abort C. Shift, move, accept, reject D. Shift, reduce, accept, error
- Q115.** For the grammar $E \rightarrow TE'$, $E' \rightarrow +TE' \mid \epsilon$, $T \rightarrow FT'$, $T' \rightarrow *FT' \mid \epsilon$, $F \rightarrow (E) \mid \text{id}$, which is true? A. It is left-recursive and cannot be parsed top-down B. It is suitable for LL(1) predictive parsing without backtracking C. It requires backtracking because FIRST and FOLLOW of E' overlap D. It is ambiguous
- Q116.** Ordering the parsers by the class of grammars they accept, from smallest to largest: A. $\text{CLR} \subset \text{LALR} \subset \text{SLR}$ B. $\text{SLR} \subset \text{LALR} \subset \text{CLR}$ C. $\text{LALR} \subset \text{SLR} \subset \text{CLR}$ D. All accept identical classes
- Q117.** If a compiler can determine the scope of a declaration by examining the program text alone, the language uses: A. Static policy and lexical scope B. Dynamic policy and lexical scope C. Static policy and dynamic scope D. Dynamic policy and dynamic scope
- Q118.** Which statement about left recursion is accurate? A. It has no effect on recursive-descent parsing B. It occurs when a non-terminal appears at the end of its own production C. It can be direct or indirect and causes infinite recursion in top-down parsers D. It is required to build LL(1) parsers
- Q119.** Which technique does a compiler use to recognise identifiers during lexical analysis? A. Context-free grammar with a pushdown stack B. A deterministic finite automaton C. Operator precedence relations D. Attribute grammars
- Q120.** Common subexpression elimination and dead code removal belong to which phase? A. Lexical analysis B. Syntax analysis C. Code optimisation D. Symbol table construction
- Q121.** In three-address code, the statement $x = a * b + c * d$ requires how many temporary variables at minimum? A. 1 B. 2 C. 3 D. 4
- Q122.** The data structure used by a compiler to store identifier names along with their type, scope and memory location is the: A. Parse tree B. Symbol table C. Activation record D. Syntax-directed definition

Section H — Discrete Mathematics & Probability (Q123–134)

- Q123.** The power set of the set $\{a, b\}$ is: A. The set containing only $\{a, b\}$ B. The set of all subsets of $\{a, b\}$ C. The set of singleton subsets only D. The empty set
- Q124.** The number of reflexive relations on a set with n elements is: A. 2^n B. 2^{n^2} C. 2^{n^2-n} D. n^2
- Q125.** The number of distinct functions from a set of 4 elements to a set of 3 elements is: A. 12 B. 64 C. 81 D. 24
- Q126.** Given $E(X) = 2$, $E(Y) = 3$ and $E(XY) = 10$, the covariance of X and Y is: A. 1 B. 4 C. 5 D. 6
- Q127.** For a binomial distribution with $n = 20$ and $p = 0.3$, the variance is: A. 4.2 B. 6.0 C. 14.0 D. 20.0
- Q128.** The cofactor expansion of a 3×3 determinant along the first row is: A. The sum of each first-row element multiplied by its corresponding cofactor B. The sum of only the diagonal elements times their

cofactors C. The sum of first-column elements times their cofactors D. The difference between first-row elements and their cofactors

Q129. Which statement is true for a Hermitian matrix? A. All entries are real B. All diagonal elements are real C. The trace is always zero D. All entries are positive

Q130. In a simple undirected graph, the sum of the degrees of all vertices equals: A. The number of edges B. Twice the number of edges C. The number of vertices D. Half the number of edges

Q131. A box contains 3 red and 5 blue balls. Two balls are drawn without replacement. The probability that both are red is: A. $\frac{3}{28}$ B. $\frac{9}{64}$ C. $\frac{3}{14}$ D. $\frac{1}{4}$

Q132. A measurement is recorded as 12.8 cm when the true value is 13.0 cm. The absolute error is: A. 0 cm B. 0.1 cm C. 0.2 cm D. 1.2 cm

Q133. Which of the following is **not** a property of a discrete random variable X taking values $0, 1, 2, \dots$? A. $P(X = x) \geq 0$ for all x B. The sum of $P(X = x)$ over all x equals 1 C. $P(X = x)$ can exceed 1 for some x D. The expectation is the sum of $x \cdot P(X = x)$

Q134. In how many ways can 5 distinct books be arranged on a shelf such that 2 particular books are always together? A. 24 B. 48 C. 60 D. 120

Section I — Programming in C (Q135–144)

Q135. Given `int a = 10; int *p; int **w; p = &a; w = &p;` which expression yields the value 10? A. `*w` B. `**w` C. `&p` D. `*p * *w`

Q136. What is the output?

```
☐
c
int a[5] = {1,2,3,4,5};
int *p = a;
printf("%d", *(p+3) + *(a+1));
```

A. 5 B. 6 C. 7 D. 8

Q137. What is printed?

```
☐
c
int i = 5;
printf("%d %d", i++, ++i);
```

A. 5 7 B. 5 6 C. 6 7 D. Undefined behaviour — the result is not specified by the standard

Q138. Which storage class retains a local variable's value between successive calls to a function? A. auto B. register C. static D. extern

Q139. Which is a valid declaration of a 2-D array in C? A. `int a[][];` B. `int a[3][];` C. `int a[][4];` as a function parameter D. `int a[,4];`

Q140. What is the output?

```
☐
c
int f(int n){ return n<=1 ? 1 : n*f(n-1); }
printf("%d", f(5));
```

A. 24 B. 60 C. 120 D. 720

Q141. On a machine where `int` is 4 bytes and pointers are 8 bytes, `sizeof(arr)` where `int arr[10];` is declared as a **local array** gives: A. 8 B. 10 C. 40 D. 80

Q142. Which is true about the relationship between a 1-D array name and a pointer in C? A. The array name is a modifiable pointer variable B. The array name decays to a pointer to its first element but is not itself assignable C. They are completely unrelated D. `sizeof` returns the same value for both in all contexts

Q143. Why is it good practice to close files explicitly even though the runtime may close them at program exit? A. To reset file permissions B. To release file descriptors promptly and flush buffers, avoiding resource leaks C. To prevent the file from being deleted D. To hide the file from other processes

Q144. What does `malloc(10 * sizeof(int))` return on failure? A. 0 cast to int B. NULL C. A pointer to an unusable block D. It raises an exception

Section J — Object-Oriented Programming concepts & Software Engineering (Q145–154)

Q145. Which pair correctly matches the OOP principle to its definition? A. Encapsulation — deriving a new class from an existing one B. Polymorphism — bundling data and methods and restricting direct access C. Abstraction — exposing essential features while hiding implementation detail D. Inheritance — one interface serving multiple underlying forms

Q146. Method overloading and method overriding are resolved respectively at: A. Run time and compile time B. Compile time and run time C. Both at compile time D. Both at run time

Q147. The diamond problem in multiple inheritance is resolved in C++ using: A. Friend functions B. Virtual base classes C. Static members D. Namespaces

Q148. Which relationship represents "a Car **has-a** Engine, and the Engine cannot exist without the Car"? A. Association B. Aggregation C. Composition D. Inheritance

- Q149.** In good software design, the desirable combination is: A. High cohesion and high coupling B. High cohesion and low coupling C. Low cohesion and low coupling D. Low cohesion and high coupling
- Q150.** Which SDLC model is most appropriate when requirements are stable, well understood and unlikely to change? A. Agile B. Spiral C. Waterfall D. Incremental
- Q151.** The primary drawback of the Waterfall model is that: A. It has no documentation B. Returning to an earlier phase after moving forward is difficult and costly C. It cannot be used for large projects D. It omits the testing phase
- Q152.** Testing performed without knowledge of the internal code structure is called: A. White box testing B. Black box testing C. Unit testing D. Regression testing
- Q153.** For a control flow graph with 12 edges and 9 nodes, the cyclomatic complexity $V(G) = E - N + 2$ equals: A. 3 B. 4 C. 5 D. 21
- Q154.** Modifying software to work on a new operating system is which type of maintenance? A. Corrective B. Adaptive C. Perfective D. Preventive

Section K — Java (Q155–166)

- Q155.** Which statement about JDK, JRE and JVM is correct? A. $JVM \supset JRE \supset JDK$ B. $JDK \supset JRE \supset JVM$ C. $JRE \supset JDK \supset JVM$ D. They are interchangeable terms
- Q156.** What is the output?

```
java
String a = "hello";
String b = "hello";
String c = new String("hello");
System.out.println((a == b) + " " + (a == c) + " " + a.equals(c));
```

- A. true true true B. true false true C. false false true D. true true false

- Q157.** Which class should be used for heavy string concatenation inside a loop in a single-threaded program? A. String B. StringBuffer C. StringBuilder D. CharSequence
- Q158.** Which of the following is a **checked** exception? A. NullPointerException B. ArrayIndexOutOfBoundsException C. IOException D. ArithmeticException

- Q159.** Which statement about `final`, `finally` and `finalize` is correct? A. `final` is a block, `finally` is a keyword, `finalize` is a class B. `final` restricts modification, `finally` always executes after try-catch, `finalize` is a method called before garbage collection C. All three are access modifiers D. `finally` is used to make a class non-inheritable

Q160. The key difference between HashMap and Hashtable is: A. HashMap is synchronised and permits null keys B. Hashtable is synchronised and permits no null key or value; HashMap is unsynchronised and permits one null key C. Hashtable is faster in all cases D. HashMap maintains insertion order

Q161. Which collection should be chosen for frequent insertions and deletions in the middle of a sequence? A. ArrayList B. LinkedList C. HashSet D. TreeMap

Q162. What is the output?

```
java
int[] arr = {1,2,3};
System.out.println(arr.length + " " + "abc".length());
```

A. 3 3 B. 4 3 C. 3 4 D. Compilation error

Q163. In Java, all classes implicitly extend: A. Class B. Object C. Serializable D. Comparable

Q164. Which order of execution is correct when an object of a subclass is created? A. Subclass constructor → superclass constructor → instance initialiser B. Static block → instance initialiser → constructor C. Constructor → static block → instance initialiser D. Instance initialiser → static block → constructor

Q165. Which method must be called to start a new thread of execution? A. `run()` B. `start()` C. `execute()` D. `init()`

Q166. From Java 8 onwards, an interface may contain: A. Only abstract methods B. Abstract, default and static methods C. Only constants D. Constructors

Section L — C++ (Q167–178)

Q167. Which is a correct difference between a pointer and a reference in C++? A. A reference can be reassigned; a pointer cannot B. A reference must be initialised at declaration and cannot be rebound; a pointer can be null and reassigned C. Both can be null D. References occupy no memory and pointers occupy none either

Q168. A base class destructor should be declared `virtual` in order to: A. Improve performance B. Ensure the derived class destructor runs when deleting through a base class pointer C. Allow multiple inheritance D. Permit operator overloading

Q169. The mechanism that enables run-time polymorphism in C++ is: A. Function overloading B. Templates C. The virtual function table (vtable) D. Friend functions

Q170. Which is true about `malloc/free` versus `new/delete`? A. `malloc` calls the constructor; `new` does not B. `new` allocates memory and calls the constructor; `malloc` only allocates raw memory C. `free` calls the destructor D. They are fully interchangeable

Q171. A copy constructor is invoked in all of the following cases **except**: A. Passing an object by value to a function B. Returning an object by value C. Initialising an object from another object of the same class D. Assigning to an already-constructed object with `=`

Q172. Which STL container provides $O(\log n)$ lookup and keeps elements sorted by key? A. `unordered_map` B. `map` C. `vector` D. `deque`

Q173. The output of the following is:



```
cpp
int x = 5;
int &r = x;
r = 10;
cout << x;
```

A. 5 B. 10 C. Address of x D. Compilation error

Q174. Which operator **cannot** be overloaded in C++? A. `+` B. `[]` C. `::` D. `<<`

Q175. RAII (Resource Acquisition Is Initialisation) primarily ensures that: A. Resources are acquired lazily B. Resources are released automatically when the owning object goes out of scope C. All allocations use `malloc` D. Destructors are never called

Q176. A class template differs from a function template in that: A. Class templates cannot have multiple type parameters B. Class templates are instantiated with explicit type arguments to create a concrete class C. Function templates cannot be overloaded D. Class templates are resolved at run time

Q177. Declaring a member function `const` means: A. The return value cannot be modified B. The function cannot modify non-mutable data members of the object C. The function can only be called once D. The object is allocated in read-only memory

Q178. In C++, if a class defines no constructor at all, the compiler: A. Reports an error B. Generates a default constructor C. Generates a parameterised constructor D. Makes the class abstract

Section M — Python (Q179–190)

Q179. Which of the following Python types is **immutable**? A. list B. dict C. set D. tuple

Q180. What is the output?



```
python
```

```
a = [1, 2, 3]
b = a
b.append(4)
print(len(a))
```

A. 3 B. 4 C. 0 D. Error

Q181. What is the output?

```
python
def f(x, lst=[]):
    lst.append(x)
    return lst
print(f(1), f(2))
```

A. `[1] [2]` B. `[1] [1, 2]` C. `[1, 2] [1, 2]` D. `[1] [2, 1]`

Q182. What does `print([i*i for i in range(5) if i % 2 == 0])` output? A. `[0, 1, 4, 9, 16]` B. `[0, 4, 16]` C. `[1, 9]` D. `[0, 2, 4]`

Q183. The difference between `is` and `==` in Python is: A. `is` compares values, `==` compares identity B. `is` compares object identity (memory address), `==` compares values C. They are identical in behaviour D. `is` works only on integers

Q184. What is the output of `print("hello"[::-1])`? A. `hello` B. `olleh` C. `h` D. Error

Q185. The Global Interpreter Lock (GIL) in CPython means: A. Only one thread executes Python bytecode at a time in a single process B. Python cannot use threads at all C. Multiprocessing is impossible D. All Python code is executed in parallel automatically

Q186. A generator function differs from a normal function because it: A. Returns a list of all values at once B. Uses `yield` to produce values lazily, one at a time, preserving state C. Cannot be iterated over D. Always runs faster

Q187. What is the output?

```
python
t = (1, 2, [3, 4])
t[2].append(5)
print(t)
```

A. `(1, 2, [3, 4])` B. `(1, 2, [3, 4, 5])` C. `TypeError` D. `(1, 2, 5)`

Q188. In a function signature `def f(*args, **kwargs):` A. `args` is a dict and `kwargs` is a tuple B. `args` is a tuple of positional arguments and `kwargs` is a dict of keyword arguments C. Both are lists D. Neither can be empty

Q189. Which correctly describes a decorator? A. A function that takes a function and returns a modified function B. A class method that cannot be overridden C. A way to declare variable types D. A built-in exception handler

Q190. What is the output?

```
python
print(list(map(lambda x: x*2, filter(lambda x: x > 2, [1, 2, 3, 4]))))
```

A. `[2, 4, 6, 8]` B. `[6, 8]` C. `[3, 4]` D. `[2, 4]`

Section N — Artificial Intelligence & Machine Learning (Q191–200)

Q191. Which of the following is an **unsupervised** learning algorithm? A. Logistic regression B. Random forest C. k-means clustering D. Support vector machine

Q192. A model that performs very well on training data but poorly on unseen test data is exhibiting: A. Underfitting B. Overfitting C. High bias D. Data leakage by definition

Q193. A binary classifier produces TP = 40, FP = 10, FN = 20, TN = 30. What is its precision? A. 0.57 B. 0.67 C. 0.75 D. 0.80

Q194. For the same confusion matrix as Q193, the recall is: A. 0.57 B. 0.67 C. 0.75 D. 0.80

Q195. Increasing model complexity generally has which effect on the bias-variance tradeoff? A. Bias increases, variance decreases B. Bias decreases, variance increases C. Both increase D. Both decrease

Q196. Which activation function suffers most severely from the vanishing gradient problem in deep networks? A. ReLU B. Leaky ReLU C. Sigmoid D. Identity

Q197. In the A* search algorithm, the evaluation function is $f(n) = g(n) + h(n)$. A* is guaranteed to find an optimal solution when $h(n)$ is: A. Always zero B. Admissible (never overestimates the true cost to the goal) C. Always greater than $g(n)$ D. Randomly chosen

Q198. Alpha-beta pruning applied to the minimax algorithm: A. Changes the final value returned by minimax B. Returns the same result as minimax while examining fewer nodes C. Works only on single-player games D. Requires the game tree to be complete

Q199. The "naive" assumption in the Naive Bayes classifier is that: A. All classes are equally likely B. Features are conditionally independent given the class label C. The data follows a uniform distribution D. There are exactly two classes

ANSWER KEY WITH EXPLANATIONS

Section A — COA / Digital Logic

Q	Ans	Explanation
1	C	2^{32} addresses $\times$ 4 bytes per word = 2^{34} bytes = 16 GB. If it were byte-addressable the answer would be 4 GB — read the wording.
2	C	Sign 0; exponent 10000011 = 131, so $e = 131 - 127 = 4$; mantissa $1.1001_2 = 1.5625$; $1.5625 \times 2^4 = 25.0$.
3	C	Offset = $\log_2 32 = 5$ bits. Blocks = 16 KB / 32 B = 512, so index = 9 bits. Tag = $32 - 5 - 9 = 18$.
4	C	Capacity ratio 1M/128K = 8 chips deep; width ratio 16/8 = 2 chips wide. $8 \times 2 = 16$.
5	B	The key register masks which bits of the argument register take part in the parallel comparison.
6	B	One rotation = $60/12000$ s = 5 ms. Average latency = half a rotation = 2.5 ms.
7	C	RAID 5 uses one disk's worth of capacity for distributed parity: $(6 - 1) \times 2$ TB = 10 TB.
8	B	Time = (number of instructions + stages - 1) $\times$ clock = $(1000 + 4) \times 2 = 2008$ ns.
9	B	$0.9 \times 10 + 0.1 \times (10 + 100) = 9 + 11 = 20$ ns.
10	B	Need $2^r \geq m + r + 1$. With $m = 8$: $r = 4$ gives $16 \geq 13$. $r = 3$ gives $8 \geq 12$, which fails.
11	A	11011011 $\rightarrow$ nibbles 1101 1011 = D B hex. Groups of three from the right: 011 011 011 = 333 octal. Both equal 219.
12	B	n-bit two's complement spans -2^{n-1} to $2^{n-1} - 1$.
13	B	$2^3 = 8 < 12 \leq 16 = 2^4$, so 4 flip-flops.
14	C	J = K = 1 is the toggle condition.
15	C	Indexed addressing adds an index register to a base address, incrementing through array elements.

Q Ans	Explanation
16 A	Cycle stealing lets the DMA controller take single bus cycles, interleaving with CPU activity rather than blocking it entirely.
17 A	Postfix $AB+ \rightarrow (A+B)$; $CD- \rightarrow (C-D)$; the final $*$ multiplies them.
18 B	Registers are fastest and smallest; secondary storage slowest and largest.
19 C	SRAM uses flip-flops (fast, expensive, no refresh); DRAM uses capacitors (cheap, dense, needs refresh).
20 C	$64 \text{ KB} = 2^{16}$ locations, so 16 address bits.
21 A	$A'B + AB' + AB = A'B + A(B' + B) = A'B + A = A + B$.
22 B	$0.95 \times 2 + 0.05 \times [0.9 \times (2+10) + 0.1 \times (2+10+100)] = 1.9 + 0.05 \times (10.8 + 11.2) = 1.9 + 1.5 = 3.4 \text{ ns}$.

Section B — Operating Systems

Q Ans	Explanation
23 B	FIFO with 3 frames faults on 1,2,3,4,1,2,5,3,4 = 9 faults (5 is a hit at position 12; 1 and 2 hit at positions 8 and 9).
24 C	FIFO with 4 frames faults 10 times — more frames, more faults.
25 B	Belady's anomaly: increasing frame count increases page faults. Only possible with non-stack algorithms like FIFO.
26 C	LRU gives 10 faults here. LRU is a stack algorithm so it never shows Belady's anomaly, but it is not always better than FIFO on a given string.
27 C	$2^{32}/2^{12} = 2^{20}$ entries $\times$ 4 bytes = 4 MB.
28 C	$0.8 \times (20 + 100) + 0.2 \times (20 + 100 + 100) = 96 + 44 = 140 \text{ ns}$.
29 B	SJF order 3, 6, 7, 8. Waiting times 0, 3, 9, 16 $\rightarrow$ total 28 / 4 = 7.00.
30 C	$2^3 = 8$ total processes exist, of which 1 is the original — so 7 children.
31 C	The four conditions are mutual exclusion, hold and wait, no preemption, and circular wait.
32 B	SSTF order: 65, 67, 37, 14, 98, 122, 124, 183. Movement = $12+2+30+23+84+24+2+59 = 236$.

Q Ans	Explanation
33 B	Fixed partitions are created at boot time and never resized, so a process smaller than its partition wastes the remainder internally.
34 A	Definition of thrashing. The fix is reducing the degree of multiprogramming.
35 B	LRU is a stack algorithm — the set of pages in k frames is always a subset of the set in k+1 frames.
36 B	<code>read()</code> traps to the kernel, which copies data from kernel buffers into the caller's user-space buffer.
37 B	A safe state has at least one safe sequence in which each process can finish with currently available plus released resources.
38 D	Associative is a memory-organisation term, not a file allocation strategy.
39 B	Preemptive scheduling moves the running process back to the ready queue (it is not terminated).
40 B	Segments correspond to logical program units of varying length; pages are uniform physical blocks.
41 B	A binary semaphore initialised to 1 acts as a mutex lock.
42 C	A blocked process must first return to the ready queue; it cannot be dispatched directly.

Section C — Computer Networks

Q Ans	Explanation
43 B	$/26$ leaves 6 host bits: $2^6 - 2 = 62$ usable addresses.
44 B	Subnets are .0–.63, .64–.127, .128–.191, .192–.255. The second subnet's broadcast is 192.168.10.127.
45 C	$C = B \log_2(1 + \text{SNR}) = 3000 \times \log_2(1024) = 3000 \times 10 = 30,000$ bps.
46 B	Go-Back-N allows $2^n - 1$ outstanding frames to avoid ambiguity.
47 C	Selective Repeat needs the window to be at most half the sequence space: 2^{n-1} .
48 C	The transport layer provides process-to-process reliability; the data link layer is hop-to-hop.
49 B	The root server does not resolve the full name; it refers the resolver to the correct TLD server.
50 B	The sender must still be transmitting when a collision signal returns, which sets a floor on frame size relative to propagation time.

Q Ans	Explanation
51 B	SMTP is 25 and DNS is 53. HTTP is 80, FTP 20/21, SSH 22, Telnet 23.
52 B	$10^9 \text{ bits/s} \times 10 \times 10^{-3} \text{ s} = 10^7 \text{ bits} = 10 \text{ Mb}$.
53 B	Connection-oriented service sets up state and guarantees ordered reliable delivery; connectionless (e.g. UDP) does not.
54 C	Routers forward based on IP addresses — network layer. Switches are data link.
55 B	Slow start doubles cwnd every RTT (exponential growth) until ssthresh, then switches to linear congestion avoidance.
56 C	IPv6 uses 128-bit addresses versus IPv4's 32.
57 A	ARP maps IP to MAC. RARP does the reverse.
58 B	Link state protocols (OSPF) flood link information so every router builds the full topology and runs Dijkstra.
59 B	$1500 \times 8 = 12,000 \text{ bits} / 10^7 \text{ bps} = 1.2 \times 10^{-3} \text{ s} = 1.2 \text{ ms}$.
60 B	Addressing, error control, flow control and sequencing recur as design issues across several layers.

Section D — Data Structures & Algorithms

Q Ans	Explanation
61 C	Singly linked = data + one next pointer. Two pointers describes a doubly linked list.
62 B	Leaving one slot empty distinguishes full from empty in the standard circular queue implementation.
63 C	Preorder gives the root; inorder splits left and right subtrees. Preorder + postorder is not sufficient for general binary trees.
64 B	Catalan number $C_4 = (1/5) \cdot C(8,4) = 70/5 = 14$.
65 B	Master theorem case 2: $a = 2, b = 2, f(n) = n = \Theta(n^{\log_b a}) \rightarrow \Theta(n \log n)$.
66 C	Bottom-up heapify is $O(n)$; inserting n elements one at a time would be $O(n \log n)$.
67 A	Nodes from $\lfloor n/2 \rfloor + 1$ to n are leaves, giving $\lfloor n/2 \rfloor$ leaves.
68 C	A maximally unbalanced partition at every step gives $T(n) = T(n-1) + O(n) = O(n^2)$.

Q Ans	Explanation
69 C	Merge sort is stable and $O(n \log n)$ worst case. Heapsort is $O(n \log n)$ but unstable; quicksort is $O(n^2)$ worst case.
70 C	$5+3 = 8$; $8-2 = 6$; $8 \times 6 = 48$.
71 B	$n(n-1)/2 = 8 \times 7 / 2 = 28$.
72 B	Dijkstra's greedy choice assumes distances never decrease once finalised; negative edges break that.
73 B	Order does not matter in an unsorted array, so swap in the last element and shrink — $O(1)$.
74 B	The hash map gives $O(1)$ lookup and the doubly linked list gives $O(1)$ reordering and eviction.
75 B	$\lfloor \log_2 1000 \rfloor + 1 = 9 + 1 = 10$.
76 C	$N(h) = N(h-1) + N(h-2) + 1$ with $N(0)=1, N(1)=2 \rightarrow N(2)=4, N(3)=7, N(4)=12$.
77 B	Insertion sort is adaptive and stable, $O(n)$ on nearly sorted data, $O(n^2)$ worst case, $O(1)$ extra space.
78 B	Expected chain length is α , so search is $O(1 + \alpha)$.
79 D	Sparse representations only pay off when most entries are zero.
80 C	0/1 knapsack has optimal substructure with overlapping subproblems — greedy fails on it.

Section E — DBMS

Q Ans	Explanation
81 A	A row is a tuple; a column is an attribute.
82 B	Entity integrity forbids NULL in any part of the primary key. Referential integrity is the foreign key rule.
83 C	Rectangle = entity, ellipse = attribute, diamond = relationship.
84 B	PROJECT is unary and removes columns, giving a vertical subset. SELECT (σ) removes rows and gives the horizontal partition.
85 B	$A \rightarrow B \rightarrow C \rightarrow D$ creates transitive dependencies on the key A, which violates 3NF. All non-key attributes are fully dependent on the single-attribute key, so 2NF holds.

Q Ans	Explanation
86 B	Durability is enforced by write-ahead logging: the redo log is on stable storage before commit is acknowledged.
87 C	Repeatable read prevents dirty and non-repeatable reads but allows phantoms; only serializable eliminates phantoms.
88 B	A schema is a named collection of objects with an owner's authorization identifier.
89 C	ALTER is DDL. UPDATE, INSERT and SELECT are DML/DQL.
90 C	Cartesian product = $10 \times 5 = 50$.
91 C	3NF eliminates transitive dependency of non-prime attributes on the key.
92 B	In a B+ tree all records live in the leaves, which are chained for efficient range scans.
93 B	The concurrency control component enforces isolation; the recovery manager handles atomicity and durability.
94 B	WHERE filters rows before grouping; HAVING filters groups after aggregation.
95 C	A candidate key not selected as primary is the alternate key.
96 B	2PL guarantees conflict serializability but not freedom from deadlock.
97 B	RAID 1 mirrors, halving usable capacity to 4 TB; each mirrored pair tolerates one failure.
98 B	BCNF requires the determinant of every non-trivial FD to be a super key.

Section F — Theory of Computation

Q Ans	Explanation
99 B	Two states: even count and odd count of a's, with b as a self-loop.
100 A	Each pass through the inner group consumes exactly two a's, preserving evenness.
101 B	a^+ means one or more; a^* would also accept the empty string.
102 B	States correspond to remainders 0, 1 and 2 modulo 3.
103 D	The subset construction can produce up to 2^n states.

Q	Ans	Explanation
104	B	$a^n b^n$ needs unbounded counting, so it is not regular, but a PDA accepts it.
105	C	Mealy output is a function of state and input; Moore output depends on state alone.
106	A	Moore $\rightarrow$ Mealy needs no extra states. The reverse direction (Mealy $\rightarrow$ Moore) may require more.
107	B	Pushdown automata accept exactly the context-free languages.
108	C	Type 0 = unrestricted, Type 1 = context-sensitive, Type 2 = context-free, Type 3 = regular.
109	D	Regular languages are closed under union, intersection, complement, concatenation and Kleene star.
110	B	The halting problem is undecidable but semi-decidable (recursively enumerable).

Section G — Compiler Design

Q	Ans	Explanation
111	A	The standard six-phase pipeline, with the symbol table and error handler spanning all phases.
112	C	"Parser" is a compiler component, not a lexical token class.
113	B	Left recursion causes infinite recursion in top-down predictive parsing and must be eliminated first.
114	D	Shift, reduce, accept and error are the four actions.
115	B	This is the standard left-recursion-eliminated expression grammar, which is LL(1).
116	B	$SLR(1) \subset LALR(1) \subset CLR(1)$ in terms of the grammars each can handle.
117	A	Static (compile-time) scope policy with lexical scoping determined by program text.
118	C	Left recursion means a non-terminal derives itself as the leftmost symbol; it can be direct or indirect.
119	B	Identifier patterns are regular, so a DFA recognises them during lexical analysis.
120	C	Both are machine-independent code optimisation techniques.
121	B	$t1 = ab; t2 = cd; x = t1 + t2$ — two temporaries.
122	B	The symbol table stores name, type, scope and storage information.

Section H — Discrete Mathematics & Probability

Q	Ans	Explanation
123	B	The power set is the set of all subsets, including $\emptyset$ and the set itself — 2^n elements.
124	C	The n diagonal pairs are forced; the remaining $n^2 - n$ pairs are free, giving $2^{(n^2-n)}$.
125	C	Each of the 4 domain elements has 3 choices: $3^4 = 81$.
126	B	$\text{Cov}(X,Y) = E(XY) - E(X)E(Y) = 10 - 6 = 4$.
127	A	Variance = $npq = 20 \times 0.3 \times 0.7 = 4.2$.
128	A	Expansion along a row multiplies each element by its cofactor and sums.
129	B	$A = A^H$ forces diagonal entries to equal their own conjugates, hence real. Off-diagonal entries may be complex.
130	B	Handshake lemma: each edge contributes 2 to the total degree.
131	A	$(3/8) \times (2/7) = 6/56 = 3/28$.
132	C	$ 12.8 - 13.0 = 0.2$ cm.
133	C	A probability mass function value can never exceed 1.
134	B	Treat the pair as one block: $4!$ arrangements $\times 2!$ internal orders = $24 \times 2 = 48$.

Section I — Programming in C

Q	Ans	Explanation
135	B	<code>w</code> points to <code>p</code> , <code>*w</code> is <code>p</code> , <code>**w</code> dereferences to <code>a</code> = 10.
136	B	<code>*(p+3)</code> = 4 and <code>*(a+1)</code> = 2, so $4 + 2 = 6$.
137	D	Modifying <code>i</code> twice without an intervening sequence point is undefined behaviour, not a fixed answer.
138	C	<code>static</code> gives a local variable static storage duration, preserving its value across calls.
139	C	All dimensions except the first must be specified; as a parameter <code>int a[][4]</code> is valid.
140	C	$5! = 120$.

Q	Ans	Explanation
141	C	<code>sizeof</code> on a local array gives total bytes: $10 \times 4 = 40$. Inside a function taking a pointer it would give 8.
142	B	The array name decays to a pointer in most expressions but is not an lvalue you can assign to.
143	B	Explicit closing frees descriptors immediately and flushes buffered writes.
144	B	<code>malloc</code> returns NULL on failure.

Section J — OOP concepts & Software Engineering

Q	Ans	Explanation
145	C	Abstraction exposes essential behaviour and hides implementation. The other three pairings are swapped.
146	B	Overloading is static (compile-time) binding; overriding is dynamic (run-time) binding.
147	B	Virtual inheritance ensures only one shared base subobject exists.
148	C	Composition is a strong has-a with dependent lifetime; aggregation is weaker.
149	B	High cohesion within a module, low coupling between modules.
150	C	Waterfall suits stable, fully specified requirements.
151	B	Waterfall is sequential with no easy backward flow once a phase is signed off.
152	B	Black box testing works purely from specification, ignoring internal structure.
153	C	$V(G) = E - N + 2 = 12 - 9 + 2 = 5$.
154	B	Adaptive maintenance responds to changes in the operating environment.

Section K — Java

Q	Ans	Explanation
155	B	JDK contains the JRE, which contains the JVM.
156	B	<code>a</code> and <code>b</code> share the string pool literal so <code>==</code> is true; <code>c</code> is a distinct heap object so <code>==</code> is false; <code>equals</code> compares content.

Q Ans	Explanation
157 C	StringBuilder is unsynchronised and faster than StringBuffer in single-threaded code.
158 C	IOException extends Exception (checked). The other three are RuntimeExceptions (unchecked).
159 B	Three unrelated things: a modifier, a block, and a method.
160 B	Hashtable is legacy and synchronised with no nulls; HashMap allows one null key and multiple null values.
161 B	LinkedList gives O(1) structural insertion/removal once positioned; ArrayList requires shifting.
162 A	Arrays expose a <code>length</code> field; String exposes a <code>length()</code> method. Both are 3.
163 B	<code>java.lang.Object</code> is the root of the class hierarchy.
164 B	Static blocks run once at class load, then instance initialisers, then the constructor body.
165 B	Calling <code>run()</code> directly executes on the current thread; <code>start()</code> creates a new one.
166 B	Java 8 added default and static methods to interfaces; Java 9 added private methods.

Section L — C++

Q Ans	Explanation
167 B	A reference is an alias bound once at initialisation; a pointer is a variable that can be null and reseated.
168 B	Without a virtual destructor, <code>delete basePtr</code> invokes only the base destructor — undefined behaviour for the derived part.
169 C	Each polymorphic class has a vtable; objects carry a vptr used to dispatch at run time.
170 B	<code>new</code> = allocation + construction; <code>malloc</code> = raw bytes only. Mixing the pairs is undefined behaviour.
171 D	Assignment to an already-constructed object calls the copy assignment operator , not the copy constructor.
172 B	<code>map</code> is a balanced BST with O(log n) operations and sorted iteration; <code>unordered_map</code> is a hash table with O(1) average.
173 B	<code>r</code> is an alias for <code>x</code> , so assigning through it changes <code>x</code> .

Q	Ans	Explanation
174 C	The scope resolution operator <code>::</code> cannot be overloaded (nor <code>.</code> , <code>.*</code> , <code>?:</code> , <code>sizeof</code>).	
175 B	The destructor releases the resource deterministically at scope exit — the basis of smart pointers.	
176 B	A class template is instantiated with type arguments at compile time to generate a concrete class.	
177 B	A const member function promises not to modify the object's non-mutable state.	
178 B	The compiler synthesises a default constructor only if no constructor is declared at all.	

Section M — Python

Q	Ans	Explanation
179 D	Tuples are immutable; list, dict and set are mutable.	
180 B	<code>b = a</code> binds another name to the same list object, so the append is visible through <code>a</code> .	
181 B	The default list is created once at function definition and persists across calls — the classic mutable default argument trap.	
182 B	Even values of <code>i</code> are 0, 2, 4 → squares 0, 4, 16.	
183 B	<code>is</code> tests identity; <code>==</code> invokes <code>__eq__</code> to compare value.	
184 B	<code>[::-1]</code> reverses the sequence.	
185 A	The GIL serialises bytecode execution within a process, which is why CPU-bound work uses multiprocessing.	
186 B	Generators yield lazily and resume where they left off, using O(1) memory.	
187 B	The tuple is immutable in its bindings, but the list it holds is mutable, so in-place mutation succeeds.	
188 B	<code>*args</code> collects extra positionals into a tuple; <code>**kwargs</code> collects keywords into a dict.	
189 A	A decorator wraps a callable and returns a replacement, usually applied with <code>@</code> syntax.	
190 B	filter keeps 3 and 4; map doubles them to 6 and 8.	

Section N — Artificial Intelligence & Machine Learning

Q	Ans	Explanation
191	C	k-means requires no labels. The other three are supervised.
192	B	Overfitting means the model has memorised training noise and generalises poorly.
193	D	Precision = $TP / (TP + FP) = 40 / 50 = 0.80$.
194	B	Recall = $TP / (TP + FN) = 40 / 60 = 0.667$.
195	B	More complex models fit training data more closely (lower bias) but are more sensitive to the sample (higher variance).
196	C	Sigmoid saturates at both ends, driving gradients toward zero; ReLU largely avoids this.
197	B	Admissibility (never overestimating) is the condition for A* optimality in tree search; consistency is needed for graph search.
198	B	Alpha-beta is a pure efficiency optimisation — the minimax value is unchanged.
199	B	Naive Bayes assumes conditional independence of features given the class.
200	B	Pooling downsamples feature maps, reducing computation and giving small translation invariance.

How to use this paper

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